

OEAS 705/805 Advanced Environmental Data Science

Fall 2026

Week 2

Dr. Jessie Turner, jturners@odu.edu

Class times: OCNPS 320, T/Th, 11:00 - 12:15

Office hours: After class T/Th,

And: OCNPS 411, M, 1:00 - 3:00



Course Housekeeping

Tools: - laptop

- git

- GitHub

- conda

- python

- Interface (Visual Studio Code / Jupyter Notebook)

- Text editor (Sublime Text or BBEdit or other)

Course Housekeeping

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We will get here this week

- conda
- python
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FAIR data principles review

Check the “FAIRness” of a dataset:

<https://www.f-uji.net/>



F-UJI is a web service to programatically assess FAIRness of research data objects at the dataset level based on the FAIRsFAIR Data Object Assessment Metrics [↗](#)

[Click here to assess a dataset](#)

Initial Data Access and Exploration

Data project setup

Thinking about whole data analysis projects and keeping them from getting disorganized

Initial Data Access and Exploration

Data project setup

Life cycle of your research projects (whether you think of them as specific "data science" projects or not):

- **Data Collection**
- **Data Cleaning**



Initial Data Access and Exploration



Initial Data Access and Exploration

- **Exploratory Data Analysis (EDA):** Iterate with data cleaning
- **Model Development:** build models and evaluate them (later in this course)
- **Model Deployment:** writing the paper, sharing on GitHub, or (we will not do this) setting up a web dashboard

Initial Data Access and Exploration

Structuring your project

Spend the time up front to organize, it will be worth it!

Initial Data Access and Exploration

Structuring your project

- README.md file that gives a brief but detailed overview of your project.
- requirements.txt or environment.yml file that lists the code packages and versions used.
- DATA
 - **Raw data**, the original data. These files should **never** be modified. Treat these files as read only to preserve their provenance. You might even opt to make this folder read-only to ensure that the original data are not modified.
 - **Clean data**, this is where you will save data that has been cleaned and is ready for analysis and model development.
 - **Results/Output**, you might
- CODE
 - **Exploratory Data Analysis**, often in the form of Jupyter notebooks
 - **Model development**, may be a combination of Jupyter notebooks and scripts
- FIGURES
 - **Figures** should have their own folder
- MANUSCRIPT / REPORT

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Initial Data Access and Exploration

Structuring your project

- FIGURES
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- MANUSCRIPT / REPORT
 - Background information
 - Notes
 - Outline of the paper
 - Drafts

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Initial Data Access and Exploration

Structuring your project

Example from my computer:

Project_NC_WQ				
Name	^	Date Modified	Size	Kind
>  CODE		Today, 9:53 AM	--	Folder
>  DATA		Today, 9:54 AM	--	Folder
>  FIGURES		Today, 9:54 AM	--	Folder
>  MANUSCRIPT		Today, 9:54 AM	--	Folder

What am I missing?

Initial Data Access and Exploration

Structuring your project

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Name	^	Date Modified	Size	Kind
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>  FIGURES		Today, 9:54 AM	--	Folder
>  MANUSCRIPT		Today, 9:54 AM	--	Folder

How to write a good README file:

<https://www.freecodecamp.org/news/how-to-write-a-good-readme-file/>

Preparing for Exploratory Data Analysis

Tools: - laptop

- git

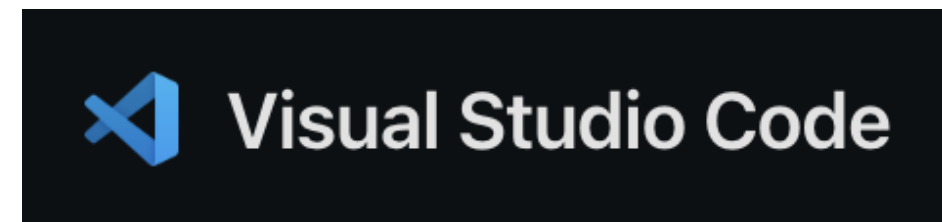
- GitHub

- Conda (Anaconda-Navigator)

- python

- Interface (Visual Studio Code
or Jupyter Notebook)

- Text editor (Sublime Text
or BBEdit
or other)



Preparing for Exploratory Data Analysis

In Terminal or PowerShell:

```
conda --version
```

```
python --version
```

Write on the board.

Also write:

your preferred interface

your preferred text editor